Image Processing and Medical Imaging

Pixel-level Methods, Histogram

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Image pre-processing

- Both input and output images are intensity images.
- Pre-processing does not increase image information. It is not feature extraction.
- The aim is an improvement of the image data, enhances some image features important for further processing, although geometric transformations of images, etc.

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Four categories of image pre-processing

- **Pixel brightness transformations (global methods)**
- Geometric transformations (see Computer Graphics)
- **Image processing using local neighborhood (local methods)**
- Image restoration (not part of this class)

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Pixel brightness transformation

- A brightness transformation modifies pixel brightness, so the transformation depends on the properties of a pixel itself. This is one of the simplest methods in image processing
- There are two classes of pixel brightness transformations: **gray-scale transformations** and **brightness corrections**.

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Gray-scale transformation

- This is a transformation T of the original brightness pixel p from scale $[p_0,p_k]$ into brightness q from a new scale $[q_0,q_k]$ is given by $q=T(p)$.
- The most common gray-scale transfromation a piecewise linear function, called brightness **tresholding**. In this case the tresholding is based on treshold values.
- It results a black-and-white (binary) image, if there are only two values in the new scale (there is only one treshold value). This is the so-called **binarization**.
- Gray-scale transformation can be supported by the hardware. Easy to realize by **look-up-table** (pallete).

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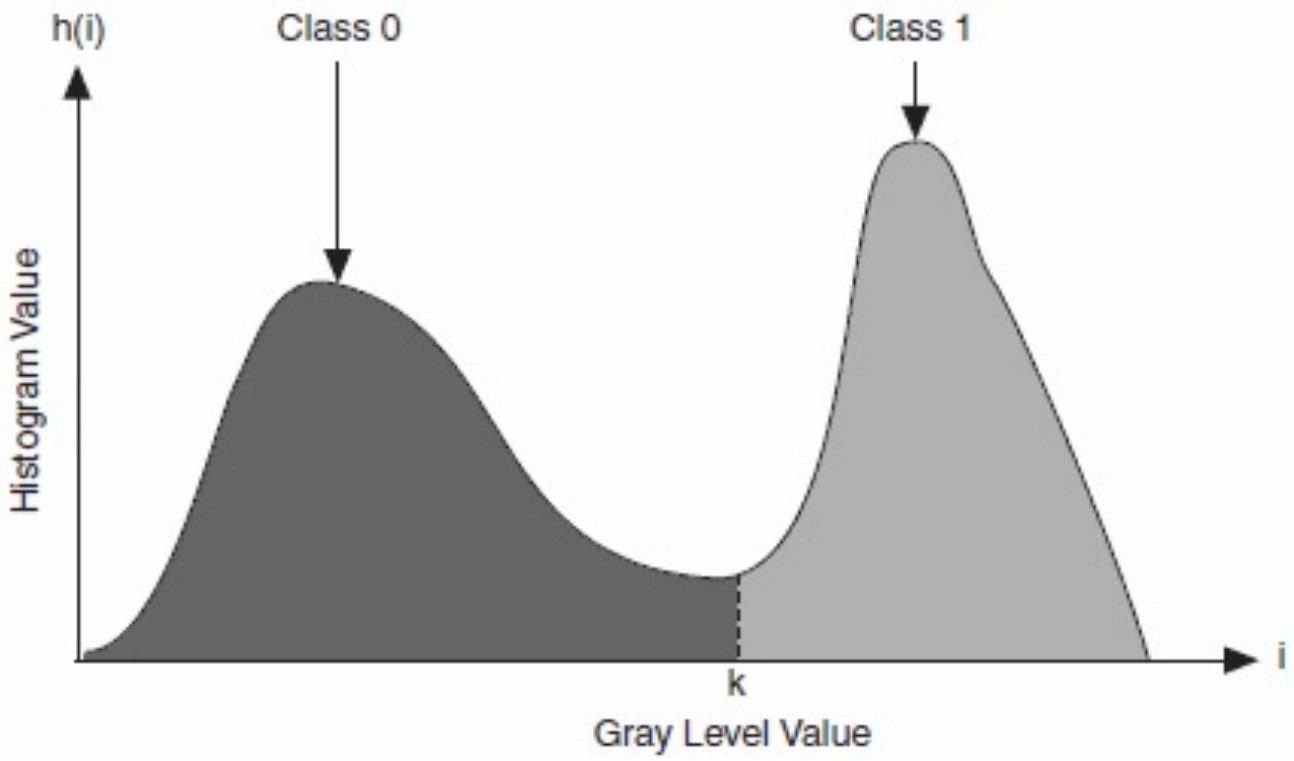
2-level tresholding with treshold value *k*



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Source: http://schorsch.efi.fh-nuernberg.de/roettger/index.php/MedicalVisualization/HistogramBasedThresholding

Example

Original image

Result

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Source: https://www.hlevkin.com/06testimages.htm

More levels (3,4 and 5-level tresholding)

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Source: Sathya P. Duraisamy, Ramanujam Kayalvizhi : A New Multilevel Thresholding Method Using Swarm Intelligence Algorithm for Image Segmentation , J. Intelligent Learning Systems & Applications, 2010, 2, 126-138

Histogram equalization

- The aim is to create an image with equally distributed brightness levels over the whole brightness scale.
- **Histogram equalization enhances contrast** for brightness values close to histogram maxima, and decreases contrast near minima.
- The (brightness) historgram of an image provides the frequency of the brightness values.
- The histogram of an image with *L* gray-levels is represented by a one-dimensional array with *L* elements.

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Histogram equalization algorithm

- For an *N x M* image of *G* gray-levels create an array *H* of length *G* initialized with 0 values.
- Form the image histogram: Scan every pixel and increment the relevant member of *H*. if pixel p has intensity g_p, perform

  $H[g_p]=H[g_p]+1$.
- Form the cumulative image histogram H_c:

  $H_c[0]=H_c[0]$,

  $H[i]=H[i-1]$, $i=1,\ldots,G-1$.

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Histogram equalization algorithm cont.

- Set $T[p]=round[(G-1)/(N \bullet M) \bullet H_c[p]$.
- Rescan the image and write an input with gray-levels g_q setting $g_q=T[g_q]$.

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Histogram equalization

Original

Equalizied

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Other gray-scale transformation transformation

- The **logarithmic gray-scale transformation** simulates the logarithmic sensitivity of the human eye to the light intensity.
- **Pseudo-color**. Since the human eye is much more sensitive to change in color than to change in brightness much more detail can be perceived in pseudo-colored images.

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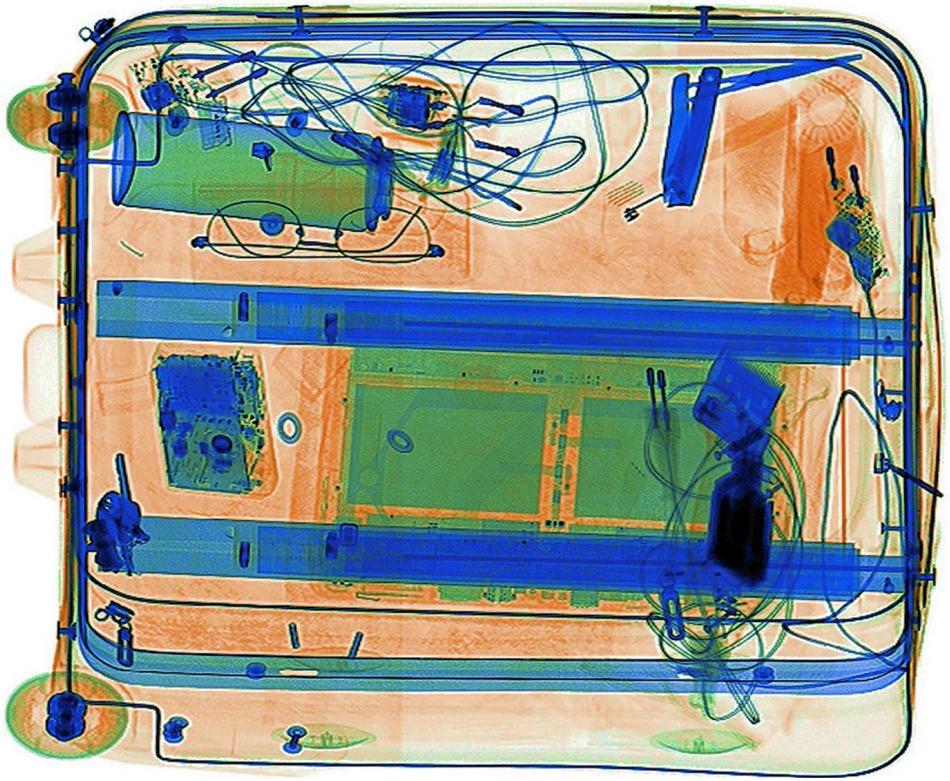
Pseudo-color image (Example)



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Source: https://www.securityprousa.com/blogs/news/x-ray-security-systems-all-you-need-to-know

Local pre-processing - filtering

- It use a small neighborhood of a pixel in an input image to produce a new brightness value of the investigated pixel in the output image. Such pre-processing operation are called also **filtering**.
- These methods can be divided into two groups: **smoothing**, **gradient operator**.

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Averaging of images (noise suppression)

- Assume that the noise value *v* at each pixel is an independent random variable with zero mean and standard deviation *s*.
- We capture the same static scene under the same conditions *n* times.
- Let us suppose the noise is normal (Gaussian).

- Let us determine $$f(i,j) = \frac{1}{n}\sum_{k=1}^{n} g_k(i,j)$$

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Averaging of images (noise suppression) cont.

- The mean value of the distribution of sample mean values is equal to the mean value of the population.
- The standard deviation of the distribution of sample mean values is

$$s / \sqrt{n}$$

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Image smoothing

- Calculation of the new value is based on the **averaging of brightness** values in some neighborhoods.
- Local image smoothing can effectively **eliminate impulse noise** or degradations apearing as thin stripes, but does not work if degradations are large blobs or thick stripes.

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Average filter

Let us determine the new value of the investigated pixel (red) based on the neighbourhood (dark blue area).

(1+4+11+2+2+5+10+5+6)/9=46/9=5.11

1	**4**	**11**	3	9	2
2	**2**	**5**	6	4	1
10	**5**	**6**	8	2	4
10	5	5	5	3	2
2	4	8	11	7	8

→

1	**4**	**11**	3	9	2
2	**5**	**5**	6	4	1
10	**5**	**6**	8	2	4
10	5	5	5	3	2
2	4	8	11	7	8

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Average filter cont.

Original

Filtered

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Median filter

Noisy image

Filtered image

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Median filter cont.

255	10	223
210	10	10
178	10	11

→

255	10	223
210	11	10
178	10	11

10	10	10	10	11	178	210	223	255

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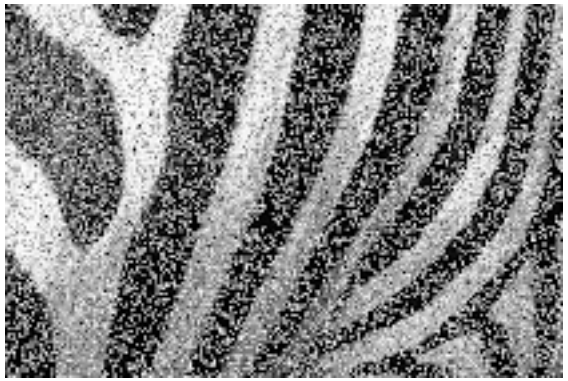
Filtering with templates (neighbourhood) of different size

Noisy image



Median filter 3x3

Median filter 5x5

Median filter 7x7

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Source:https://www.imageeprocessing.com/2017/10/part-2-2d-median-filter-with-different.html

Filtering

- Running window. Scanning pixel by pixel, row by row.
- There are two image planes in the case of sequential process.
- The size of the neighbourhood. Larger neighbourhood has stronger blur effect.
- The number of the iterations. More repetitions more blur.
- The final result without stopping? An image with one color.

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Thank you for your attention